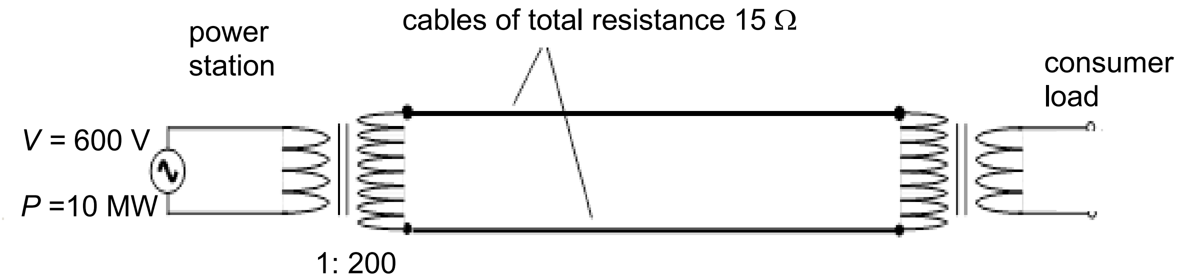
27 A 10 MW nuclear power station produces electrical power at 600 V. It uses a step-up transformer with a turns ratio of 1: 200 to increase the voltage before transmitting it over long-distance cables of total resistance 15 Ω. At the consumer load, a second transformer steps down the voltage. You may assume the transfomers are ideal. What is the power lost as heat in the cables?



A 50 kW **B** 100 kW **C** 1.0 MW **D** 960 MW

Ans: B

At the step up transformer,

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

$$\frac{600}{V_s} = \frac{1}{200}$$

$$V_s = 1.2 \times 10^5 \text{ V}$$

For ideal transfomer,

power input = power output

$$10 \times 10^6 = I_s(1.2 \times 10^5)$$

$$I_s = 83.3 \text{ A}$$

Power lost in cables,

$$P = I_s^2 R$$

$$= (83.3)^2(15)$$

$$= 1.0 \times 10^5 \text{ W}$$

$$= 100 \text{ kW}$$